Triangles & Congruence Error Analysis TEACHER KEY

Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

1. Triangle angle sum

ORIGINAL PROBLEM

The angles of a triangle measure 2x°, 3x° and 4x°. Find x.

INCORRECT WORKED SOLUTION

2x + 3x + 4x = 360

9x = 360

x = 40

A. EXPLAIN THE MISTAKE (1 POINT)

The angles of a triangle add to 180°, not 360°.

B. SHOW CORRECT WORK (1 POINT)

2x + 3x + 4x = 180

9x = 180

x = 20

Angles: 40°, 60°, 80°

2. Exterior angle

ORIGINAL PROBLEM

An exterior angle of a triangle measures 120°. The two remote interior angles measure x° and 2x°. Find x.

INCORRECT WORKED SOLUTION

x + 2x + 120 = 180

3x = 60

x = 20

A. EXPLAIN THE MISTAKE (1 POINT)

The Exterior Angle Theorem says the exterior angle EQUALS the sum of the two remote interior angles: x + 2x = 120.

B. SHOW CORRECT WORK (1 POINT)

x + 2x = 120

3x = 120

x = 40

Remote interior angles: 40° and 80°

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3. Isosceles triangle

ORIGINAL PROBLEM

An isosceles triangle has a vertex angle of 40°. Find each base angle.

INCORRECT WORKED SOLUTION

$180° - 40° = 140°$

Each base angle is 140°.

A. EXPLAIN THE MISTAKE (1 POINT)

140° is the TOTAL of the two base angles. Base angles are congruent, so divide by 2: 70° each.

B. SHOW CORRECT WORK (1 POINT)

$180° - 40° = 140°$ (sum of base angles)

Base angles are equal: $\frac{140}{2} = 70°$

Each base angle is 70°.

4. Triangle inequality

ORIGINAL PROBLEM

Can segments of length 4, 5 and 10 form a triangle?

INCORRECT WORKED SOLUTION

$5 + 10 > 4$ ✓

So yes, they form a triangle.

A. EXPLAIN THE MISTAKE (1 POINT)

The sum of EVERY pair of sides must exceed the third side. Check the two shorter sides against the longest: 4 + 5 = 9, which is less than 10.

B. SHOW CORRECT WORK (1 POINT)

Check the two shorter sides: $4 + 5 > 10$?

$9 > 10$ is false.

No, these lengths cannot form a triangle.

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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5. Possible third sides

ORIGINAL PROBLEM

Two sides of a triangle are 7 and 12. What lengths are possible for the third side?

INCORRECT WORKED SOLUTION

$7 + 12 = 19$

$12 - 7 = 5$

The third side is 5 or 19.

A. EXPLAIN THE MISTAKE (1 POINT)

The third side must be BETWEEN the difference and the sum, not equal to either: $5 < x < 19$.

B. SHOW CORRECT WORK (1 POINT)

$x < 7 + 12 = 19$

$x > 12 - 7 = 5$

$5 < x < 19$

6. Congruence shortcuts

ORIGINAL PROBLEM

$\triangle ABC$ and $\triangle DEF$ have $AB \cong DE$, $BC \cong EF$ and $\angle A \cong \angle D$. Are the triangles congruent?

INCORRECT WORKED SOLUTION

Two pairs of sides and a pair of angles are congruent.

The triangles are congruent by SSA.

A. EXPLAIN THE MISTAKE (1 POINT)

SSA is NOT a congruence rule. For SAS the angle must be BETWEEN the two sides, and $\angle A$ is not between AB and BC.

B. SHOW CORRECT WORK (1 POINT)

$\angle A$ is not the included angle for sides AB and BC.

SSA does not prove congruence.

Not enough information to conclude congruence.

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7. Corresponding parts

ORIGINAL PROBLEM

△PQR ≅ △XYZ. XZ = 9, YZ = 15 and QR = 2x + 3. Find x.

INCORRECT WORKED SOLUTION

QR ≅ XZ

2x + 3 = 9

x = 3

A. EXPLAIN THE MISTAKE (1 POINT)

Corresponding parts follow the ORDER of the letters: Q ↔ Y and R ↔ Z, so QR ↔ YZ (not XZ).

B. SHOW CORRECT WORK (1 POINT)

P ↔ X, Q ↔ Y, R ↔ Z

QR ≅ YZ

2x + 3 = 15

x = 6

8. CPCTC

ORIGINAL PROBLEM

Given: M is the midpoint of AD and of BC.
Prove: AB ≅ DC.

INCORRECT WORKED SOLUTION

1. AM ≅ DM (Definition of midpoint)
2. BM ≅ CM (Definition of midpoint)
3. AB ≅ DC (CPCTC)

A. EXPLAIN THE MISTAKE (1 POINT)

CPCTC can only be used AFTER the triangles are proven congruent. Steps are missing: ∠AMB ≅ ∠DMC (vertical angles), then △AMB ≅ △DMC (SAS).

B. SHOW CORRECT WORK (1 POINT)

1. AM ≅ DM (Midpoint)
2. ∠AMB ≅ ∠DMC (Vertical angles)
3. BM ≅ CM (Midpoint)
4. △AMB ≅ △DMC (SAS)
5. AB ≅ DC (CPCTC)

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9. AAA

ORIGINAL PROBLEM

In △ABC and △DEF, ∠A ≅ ∠D, ∠B ≅ ∠E and ∠C ≅ ∠F. Are the triangles congruent?

INCORRECT WORKED SOLUTION

All three pairs of angles are congruent.

The triangles are congruent by AAA.

A. EXPLAIN THE MISTAKE (1 POINT)

AAA only shows the triangles have the same SHAPE (they are similar). They can be different sizes, so AAA does not prove congruence.

B. SHOW CORRECT WORK (1 POINT)

AAA is not a congruence rule.

A small and a large equilateral triangle both have three 60° angles.

Not necessarily congruent (the triangles are similar).

10. Isosceles perimeter

ORIGINAL PROBLEM

An isosceles triangle has base 8 and perimeter 30. Find the length of each leg.

INCORRECT WORKED SOLUTION

Legs = 30 − 8

Each leg = 22

A. EXPLAIN THE MISTAKE (1 POINT)

30 − 8 = 22 is the total of BOTH legs. The legs are equal, so each is $\frac{22}{2} = 11$.

B. SHOW CORRECT WORK (1 POINT)

Both legs together: 30 − 8 = 22

Each leg: $\frac{22}{2} = 11$

Check: 11 + 11 + 8 = 30 ✓

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points